



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 17

TITLE : Write a Java program to demonstrate the use of Swing and Applets

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

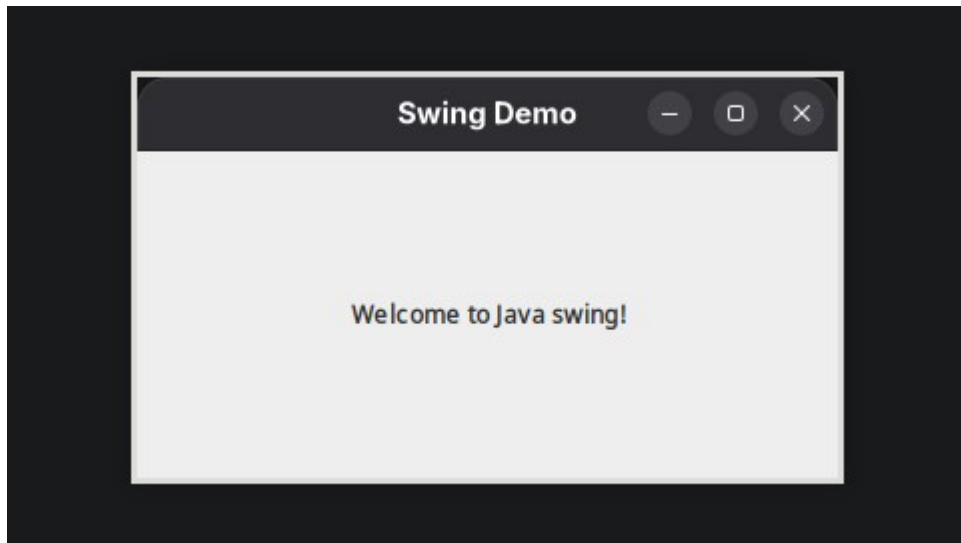
Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Theory

Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.

Output Screenshots



CODE 1

```
1 package Exp17;
2 import javax.swing.*;
3 public class Main {
4     public static void main (String[] args) {
5         JFrame frame = new JFrame("Swing Demo");
6         JLabel label = new JLabel("Welcome to Java swing!",
JLabel.CENTER);
7         frame.add(label);
8
9         frame.setSize(350, 200);
10        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
11        frame.setVisible(true);
12    }
13 }
```

CODE 2

```
1 package Exp17;
2
3 import java.applet.*;
4 import java.awt.*;
5
6 public class Main2 extends Applet {
7     public void paint (Graphics g) {
8         g.drawString("Welcome to Java Applet", 50, 50);
9     }
10 }
```

EXERCISE 1

```
1 package Exp17.Exercise;
2
3 import javax.swing.*;
4
5 public class Exp1 {
6     public static void main (String[] args) {
7         JFrame frame = new JFrame("Student Registration Form");
8
9         JLabel nameLabel = new JLabel("Name:");
10        JLabel rollLabel = new JLabel("Roll No:");
11        JLabel courseLabel = new JLabel("Course:");
12
13        JTextField nameField = new JTextField();
14        JTextField rollField = new JTextField();
15        JTextField courseField = new JTextField();
16
17        JButton submitButton = new JButton("Submit");
18
19        nameLabel.setBounds(50, 50, 100, 30);
```

```
20     nameField.setBounds(150, 50, 200, 30);
21
22     rollLabel.setBounds(50, 100, 100, 30);
23     rollField.setBounds(150, 100, 200, 30);
24
25     courseLabel.setBounds(50, 150, 100, 30);
26     courseField.setBounds(150, 150, 200, 30);
27
28     submitButton.setBounds(150, 210, 100, 30);
29
30     frame.add(nameLabel);
31     frame.add(nameField);
32     frame.add(rollLabel);
33     frame.add(rollField);
34     frame.add(courseLabel);
35     frame.add(courseField);
36     frame.add(submitButton);
37
38     frame.setSize(420, 320);
39     frame.setLayout(null);
40     frame.setVisible(true);
41
42     submitButton.addActionListener(e -> {
43         JOptionPane.showMessageDialog(
44             frame,
45             "Registration Successful!\nName: " +
nameField.getText()
46             + "\nRoll No: " + rollField.getText()
47             + "\nCourse: " + courseField.getText()
48             );
49     });
50 }
51 }
```

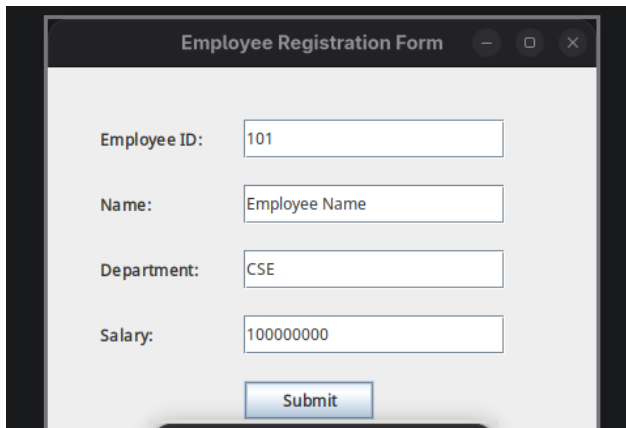
EXERCISE 2

```
1 package Exp17.Exercise;
2
3 import javax.swing.*;
4
5 public class Exp2 {
6     public static void main (String[] args) {
7         JFrame frame = new JFrame("Employee Registration
Form");
8
9         JLabel idLabel = new JLabel("Employee ID:");
10        JLabel nameLabel = new JLabel("Name:");
11        JLabel deptLabel = new JLabel("Department:");
12        JLabel salaryLabel = new JLabel("Salary:");
13
14        JTextField idField = new JTextField();
15        JTextField nameField = new JTextField();
16        JTextField deptField = new JTextField();
17        JTextField salaryField = new JTextField();
18
19        JButton submitButton = new JButton("Submit");
20
21        idLabel.setBounds(40, 40, 100, 30);
22        idField.setBounds(150, 40, 200, 30);
23
24        nameLabel.setBounds(40, 90, 100, 30);
25        nameField.setBounds(150, 90, 200, 30);
26
27        deptLabel.setBounds(40, 140, 100, 30);
28        deptField.setBounds(150, 140, 200, 30);
29
30        salaryLabel.setBounds(40, 190, 100, 30);
31        salaryField.setBounds(150, 190, 200, 30);
32
```

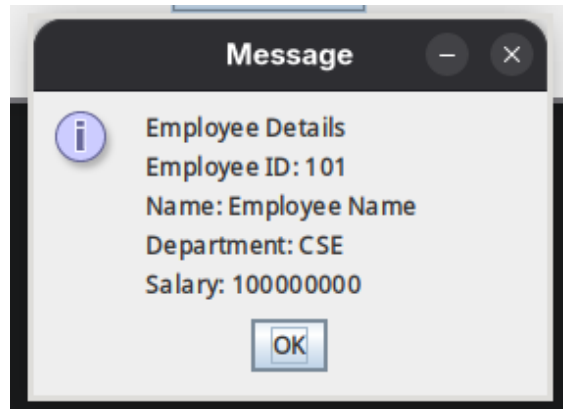
```
33     submitButton.setBounds(150, 240, 100, 30);
34
35     frame.add(idLabel);
36     frame.add(idField);
37     frame.add(nameLabel);
38     frame.add(nameField);
39     frame.add(deptLabel);
40     frame.add(deptField);
41     frame.add(salaryLabel);
42     frame.add(salaryField);
43     frame.add(submitButton);
44
45     frame.setSize(420, 350);
46     frame.setLayout(null);
47     frame.setVisible(true);
48
49     submitButton.addActionListener(e -> {
50         JOptionPane.showMessageDialog(
51             frame,
52             "Employee Details\n"
53             + "Employee ID: " + idField.getText()
54             + "\nName: " + nameField.getText()
55             + "\nDepartment: " + deptField.getText()
56             + "\nSalary: " + salaryField.getText()
57         );
58     });
59 }
60 }
```

Exercise

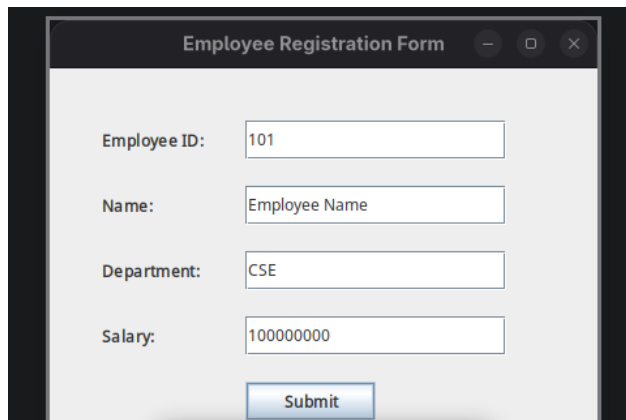
Q1. Create a simple Student Registration Form GUI using Swing.



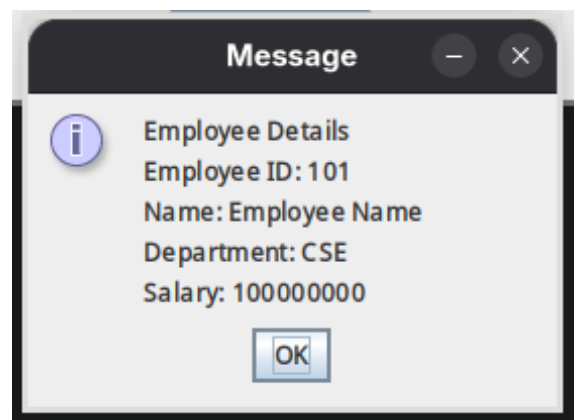
The image shows a Java Swing window titled "Employee Registration Form". It contains four text input fields with labels: "Employee ID:" (value: 101), "Name:" (value: Employee Name), "Department:" (value: CSE), and "Salary:" (value: 100000000). Below the fields is a "Submit" button.



Q2. Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box



The image shows a Java Swing window titled "Employee Registration Form". It contains four text input fields with labels: "Employee ID:" (value: 101), "Name:" (value: Employee Name), "Department:" (value: CSE), and "Salary:" (value: 100000000). Below the fields is a "Submit" button.



Conclusion

This experiment helped in understanding the use of Java Swing for creating graphical user interfaces. A simple Employee Registration Form was developed using components such as `JFrame`, `JLabel`, `JTextField`, `JButton`, and `JOptionPane`. The entered employee details were successfully collected and displayed in a dialog box, demonstrating basic GUI design and event handling in Java.